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Note

Sensitivity Analysis of DP4+ with the Probability Distribution Terms: Development of a Universal and Customizable Method

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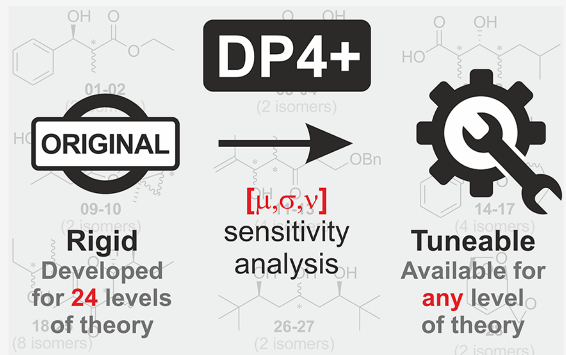
ABSTRACT: DP4+ is a popular tool for structural elucidation using GIAO NMR calculations. The method was built with 16 statistical parameters $[\mu,\sigma,\nu]$, which depend on the level of theory. Herein, we deeply analyzed the sensitivity of DP4+ when using improper $[\mu,\sigma,\nu]$ sets, a common situation found in the literature. The results led us to develop a customizable DP4+ methodology that allows preliminary calculations at any desired level of theory using a small set of training molecules.



The structural determination of molecules is of paramount importance in modern organic chemistry. This is often carried out with the aid of NMR spectroscopy,[1] which has experienced significant advances in recent decades.[2] However, the indirect nature of NMR coupled with different sources of errors and ambiguities makes the task highly challenging. In fact, the number of natural and synthetic products being incorrectly assigned from their NMR data is constantly growing.[3] Recent years have witnessed an impressive advance in the use of quantum calculations of NMR properties to facilitate the analysis.[4] Since seminal publications in the mid-2000s,[5] the discipline has been consolidated as one of the preferred alternatives for 3D elucidation.[6] Nowadays, the results of such calculations are regularly reported as valuable complements to the process of discovery of new natural and synthetic products.[7]

There are conceptually different approaches to compute the NMR properties and to correlate them with the experimental data,[8] including CASE-3D,[9] DU8+,[10] DP4,[11] DP4.2,[12] IA-DP4,[13] and *J*-DP4.[14] By late 2015, we introduced DP4+, an updated version of Goodman's DP4, which quickly became one of the most popular alternatives due to its good performance and ease of use.[15] To date, more than 200 natural products have been assigned with the aid of DP4+.[4b,16]

In this method, the probability that candidate i is correct, $P(i)$, is computed from Bayes's theorem using six cumulative t distribution functions which reflect the probability associated with the scaled and unscaled ^{1}H and ^{13}C errors e (differences between experimental and calculated chemical shifts). While scaling procedure removes systematic errors during the calculations, the inclusion of unscaled chemical shifts leads to an improved stereochemical differentiation.[15] These unscaled errors follow different t distributions depending on the hybridization of the carbon, leading to two series: sp^3 and $sp{-}sp^2$, both for proton and carbon data. Each t distribution is defined by three parameters: μ (mean), σ (standard deviation), and ν (degrees of freedom), which depend on the level of theory employed for the NMR calculation procedure. Hence, to build the DP4+ probability, 16 different statistical parameters must be known, including $[\sigma,\nu]$ for scaled ^{1}H and ^{13}C data (note that $\mu = 0$ as a consequence of the scaling) and $[\mu,\sigma,\nu]$ for unscaled ^{1}H and ^{13}C data of sp^3 and $sp{-}sp^2$ nuclei. To accomplish this task, in the original publication, the $[\mu,\sigma,\nu]$ sets were estimated at 24 levels of theory using a set of 72 known molecules featuring structural and conformational diversity.[15] The variation of those parameters with the level of theory can be noteworthy (Figure 1), which was corroborated in this study at 28 new levels of theory using a reduced test set of 8 selected molecules (vide infra). As shown in the Supporting Information, the most influential variables are the functional and basis set used in the NMR calculation step.

Although DP4+ should be employed at any of those selected 24 levels, after a recent bibliographic search, we noticed that the range of levels is much wider. Whereas in some cases the changes are minor (for instance, using B3LYP/6-31G** optimized geometries instead B3LYP/6-31G*),[17] in others,

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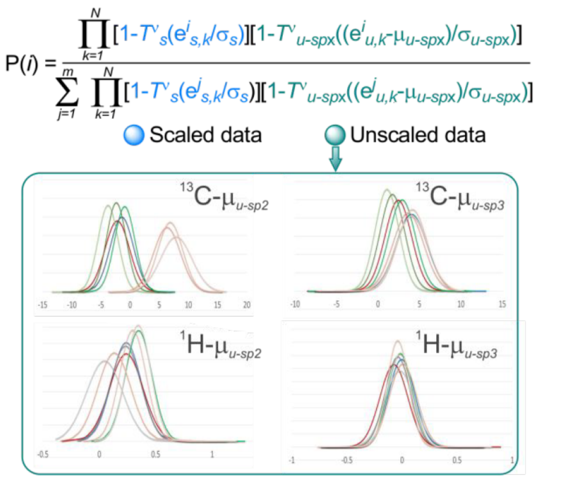

$$P(i) = \frac{\prod_{k=1}^{N}[1-T^{v}{}_{s}(e^{i}_{s,k}/\sigma_s)][1-T^{v}{}_{u\text{-}spx}((e^{i}_{u,k}-\mu_{u\text{-}spx})/\sigma_{u\text{-}spx})]}{\sum_{j=1}^{m}\prod_{k=1}^{N}[1-T^{v}{}_{s}(e^{j}_{s,k}/\sigma_s)][1-T^{v}{}_{u\text{-}spx}((e^{j}_{u,k}-\mu_{u\text{-}spx})/\sigma_{u\text{-}spx})]}$$

Figure 1. General equation for the DP4+ probability and effect of the level of theory in the μ values of the distributions of carbon and proton unscaled chemical shifts. For more detailed information about the DP4+ architecture, see ref 15.

the modifications are larger, such as including different functionals or basis sets for the GIAO NMR calculations.[18] It can be easily shown that the probability of obtaining a given error strongly depends on the $[\mu,\sigma,\nu]$ triad employed.[19] Hence, dramatic changes in the absolute probabilities computed for each candidate could be expected (numerator of the equation shown in Figure 1) using improper $[\mu,\sigma,\nu]$ sets (herein defined $[\mu,\sigma,\nu]^{\ddagger}$). However, DP4+ is relative in nature as the percentage probabilities depend on the probability values computed for all candidate isomers. Using $[\mu,\sigma,\nu]^{\ddagger}$ sets would affect the absolute probabilities of both the correct and incorrect structures with an unclear impact in the resulting stereochemical differentiation.

In order to understand the sensitivity of DP4+ with $[\mu,\sigma,\nu]$, we recomputed the DP4+ values of the same test set of 48 examples previously evaluated in the original publication (Figure 2)[15] but now correlating the chemical shifts obtained at the recommended PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* with the $[\mu,\sigma,\nu]$ sets corresponding to the other 23 levels of theory.

The best performance was achieved with the matched $[\mu,\sigma,\nu]_0$ set (Figure 3a and Table S3). The drop in the classification capacity ranged from modest (for example, see entries 2, 7–9, 15, and 20) to significant (for example, see entries 4–6 and 16–18). Up to 10 compounds (21% of the total) correctly assigned by DP4+ at the recommended level would have been misassigned using $[\mu,\sigma,\nu]^{\ddagger}$. Naturally, not all compounds exhibited such high sensitivity toward $[\mu,\sigma,\nu]$. Whereas 27 systems displayed only minor changes in the recomputed DP4+ values, the remaining 21 cases showed important variations with at least one $[\mu,\sigma,\nu]^{\ddagger}$ set.

This reinforces the fact that DP4+ should be computed using the $[\mu,\sigma,\nu]_0$ values corresponding to the level of theory used for the NMR calculations. However, considering the overwhelming number of possible combinations of functionals, basis sets, and solvent models, the development of reparametrized versions of DP4+ with enough flexibility to include most levels known would require an intractable computational effort.[20]

This raised the need to deeply understand the sensitivity of DP4+ toward each of its 16 parameters. We created an in-

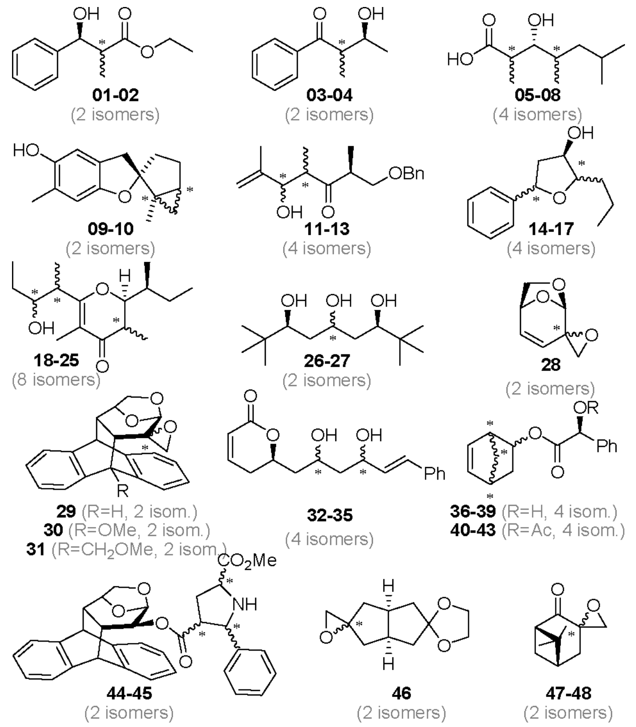

Figure 2. Test set of molecules used to evaluate the performance of DP4+ in the original publication (see ref 15).

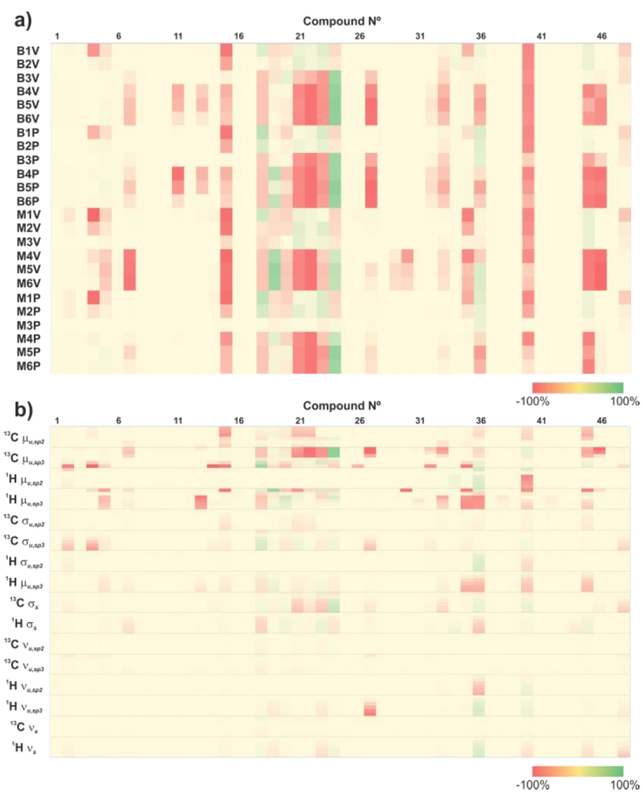

Figure 3. (a) Δ-DP4+ (DP4+$_{[\mu,\sigma,\nu]^{\ddagger}}$ - DP4+$_{[\mu,\sigma,\nu]_0}$) values computed for the test set by correlating the GIAO NMR chemical shifts calculated at the PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* level of theory, with the $[\mu,\sigma,\nu]$ parameters corresponding to different levels of theory. The codes of the levels of theory are given in the Supporting Information. (b) Dependence of DP4+ with each of its 16 $[\mu,\sigma,\nu]$ parameters. Yellowish colors indicate small variations.

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house Matlab script that recoputed DP4+ at the PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* level by keeping fixed 15 parameters at the right value corresponding to that level of theory and varying the remaining term between 6 levels within a range. The ranges were determined considering the minimum and maximum values of each parameter at all levels of theory studied in the original publication (24 levels) and in this work (28 levels). After computing the resulting >4600 DP4+ probabilities, the Δ-DP4+ values were calculated by subtracting the values to those corresponding to the DP4+ value obtained with $[\mu,\sigma,\nu]_0$. The most sensible parameters are the means of the ^{13}C and ^{1}H unscaled sp^3 series (^{13}C-μ_{u,sp^3} and ^{1}H-μ_{u,sp^3}), with averaged root-mean-square (RMS) values of 14.4 and 9.5%, respectively (Figure 3b).

Interestingly, the corresponding parameters of the sp^2 series (^{13}C-μ_{u,sp^2} and ^{1}H-μ_{u,sp^2}) were much less influential (RMS of 3.0 and 1.5%, respectively), despite having a wider range between minimum and maximum values. This might be due to the fact that the number of sp^3 carbons (or protons attached to sp^3 carbons) is larger in the molecules of the test set and to the fact that sp^3 nuclei are often more important than sp^2 in terms of stereochemical discrimination. On the other hand, the σ terms showed a more modest, yet still important, impact. Here again, the DP4+ probabilities turned out to be more sensitive, with the parameters corresponding to the sp^3 series (^{13}C-σ_{u,sp^3} and ^{1}H-σ_{u,sp^3}), than those to the sp^2 series (RMS of 2.8 vs 0.6% for carbon data and 2.6 vs 0.5% for proton data, respectively). Finally, the degrees of freedom of the t series reflected a minor impact on the resulting DP4+ probabilities, with an averaged RMS of 0.7%. This suggests that in most systems an overly accurate estimate of this term would not be mandatory.

While it was important to distinguish the role of each parameter in the DP4+ ratios, assessing their combined contribution was critical. Thus, we modified our Matlab script to compute DP4+ by correlating the NMR chemical shifts calculated at the PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* level with randomly generated $[\mu,\sigma,\nu]_r$ values by allowing the original parameters $[\mu,\sigma,\nu]_0$ to vary freely within the specified ranges discussed above. After 10,000 iterations for each compound, the differences between the DP4+ probabilities computed with $[\mu,\sigma,\nu]_0$ and $[\mu,\sigma,\nu]_r$ were significant, with averaged RMS of 25.4%. The Δ-DP4+ values were mostly negative (reddish colors in Figure 4), indicating a drop in the classification capacity. These changes impacted the sense of the assignment in 22% of the cases by reversing the most likely isomer, mostly for the worse. As previously found, not all compounds showed the same dependence with $[\mu,\sigma,\nu]$. In general, the most affected systems were those with intermediate DP4+ values computed with the matched $[\mu,\sigma,\nu]_0$ set. This is because there is no clearly identified candidate, and small perturbations in the probability distributions may end up improving the DP4+ values of another isomer. On the other hand, compounds assigned in high confidence (>99.9%) are much less prone to experience significant drifts with changes in $[\mu,\sigma,\nu]$.

In order to determine the effect of minor changes in the $[\mu,\sigma,\nu]$ set, we repeated the calculations shown in Figure 4a but using much smaller ranges. The ranges were estimated from the differences between the $[\mu,\sigma,\nu]$ parameters obtained with the full set of 72 molecules employed in the original publication with the corresponding values estimated with a reduced set of 8 selected molecules (vide infra). After 10,000 iterations for each compound in the test set, we noticed minor

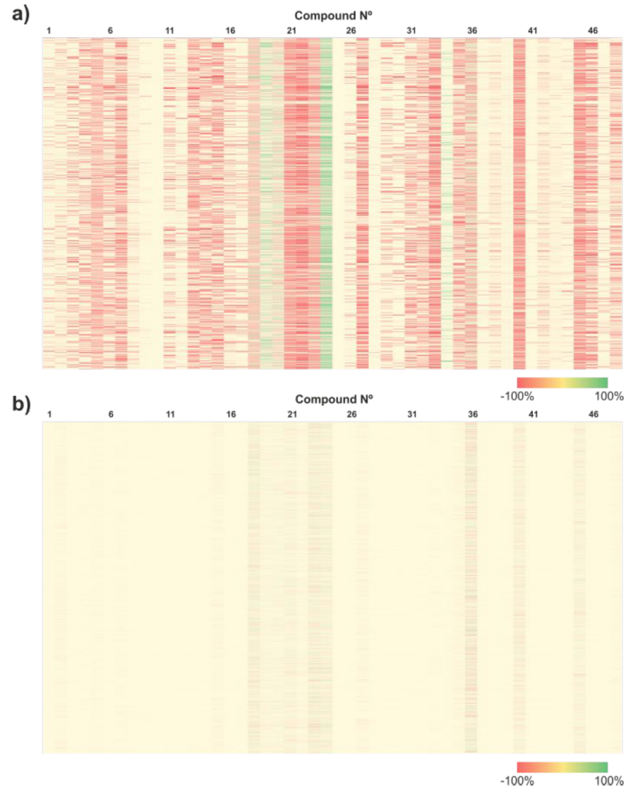

Figure 4. Δ-DP4+ (DP4+$_{[\mu,\sigma,\nu]^{\ddagger}}$ − DP4+$_{[\mu,\sigma,\nu]_0}$) values computed for the test set by correlating the GIAO NMR chemical shifts calculated at the PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* level of theory, with randomly generated $[\mu,\sigma,\nu]$ parameters by allowing them to vary within wide ranges (a) and small ranges (b). Yellowish colors indicate small variations.

Δ-DP4+ values (Figure 4b), with averaged RMS of only 2%. In fact, in less than 1% of the cases, the sense of the assignment was modified, mainly in those systems with no clear preference toward any particular isomer.

This finding allowed the generation of a customizable user-defined DP4+ method. In this approach, the DP4+ code is open for the modification of the $[\mu,\sigma,\nu]$ values, which in turn should be estimated for any suitable level of theory using a large set of known molecules (for example, those 77 examples provided in the original publication). In order to be easy to handle for the end-user, the new DP4+ was coded as an Excel file which can be downloaded for free from http://www.sarotti-NMR.weebly.com or http://www.iquir-conicet.gov.ar/Sarotti-NMR. This has the extra advantage of being available for up to 128 isomers and 300 chemical shifts and having incorporated four additional spreadsheets to visualize the unscaled and scaled chemical shifts and the corresponding differences with the experimental values. This helps analyze the overall agreement between experimental and calculated data for all isomers and the rapid identification of outliers, as well. A full instructive for its use is provided in the Supporting Information.

The development of customizable versions of DP4+ at new levels of theory requires, as has been said, the adequate determination of the 16 statistical parameters at the chosen level using a large set of known molecules. Although this would be the recommended procedure, it is somewhat cumbersome, especially in the early stages of methodological exploration. Considering the low relative sensitivity that DP4+ showed to

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minor variations of [μ,σ,ν], we speculated that a reasonably certain estimation of these parameters using a smaller set of molecules would allow obtaining DP4+ results close enough the expected values but at remarkably lower cost. Given that the most sensitive terms are μ and σ, we next sought to find a representative set composed of a few molecules that would provide [μ,σ] parameters as close as possible to those obtained with a much larger set at all levels of theory. We also foresaw that these molecules should be well-known and rigid, so that their NMR properties could be estimated without the need for expensive conformational searches. To that end, we used an in-house Matlab script that explored millions of combinations of molecules within a predefined set that fulfilled the initial requirements. The best match was obtained with the eight molecules shown in Figure 5, with an averaged error of 6 and 12% for the μ and σ terms, respectively. However, using such a small set precluded even a rough estimation of the ν terms. Since those 6 parameters were identified as the least sensitive ones, and considering that t distributions with ν > 20 show remarkable similarity with normal distributions, we hypothesized that using the averaged values over the 24 levels of theory would afford a reasonable guess. As expected, the use of estimated [μ,σ,ν] terms had a minor effect on the resulting DP4+ values. The averaged Δ-DP4+ for the full test set was only 2.3% at the PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* level, with a maximum Δ-DP4+ value of 17.8%. In none of the cases was the sense of the assignment affected. As expected, using the exact [ν] values corresponding to the corresponding level of theory hardly affected the results (with Δ-DP4+ of 1.7%, and Max[Δ-DP4+] of 14.3%). These results demonstrated that a rough estimation of the [μ,σ,ν] allows excellent results for preliminary trials, though more refined values should be used for more refined applications.

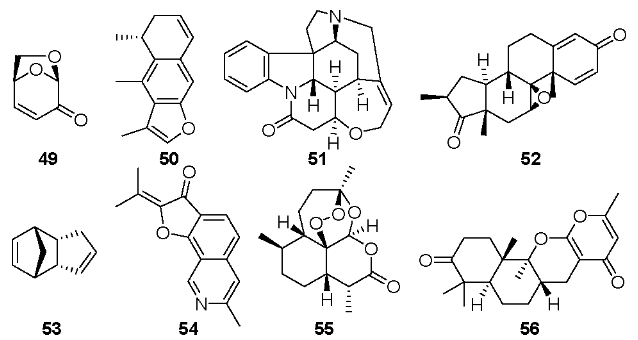

Figure 5. Representative set of molecules that leads to good estimations of the [μ,σ] terms.

The results collected in this work reinforce the recommendation of computing DP4+ by correlating GIAO NMR chemical shifts with [μ,σ,ν] parameters obtained at the same level of theory. Otherwise, potential misassignments might take place whenever the discrepancies between the [μ,σ,ν] actually employed and those that should have been used are large. On the contrary, the DP4+ results showed subtle dependency with small Δ-[μ,σ,ν] values. This allowed the generation of a tunable method, in which the DP4+ probabilities can be computed at any desired level of theory. The [μ,σ] terms can be fairly estimated from a small set of 8 rigid molecules for further preliminary explorations. However, if more accurate DP4+ results are required, the [μ,σ,ν] parameters should be obtained from a much more exhaustive number of molecules, such as those employed in the original publication.

Finally, we wish to provide a general guidance for DP4+ practitioners. Besides the caution of using the proper [μ,σ,ν] values, it should also be emphasized that the results would rely on the quality of the data employed. Incorrectly assigned NMR signals and/or improper computational work might lead to erroneous results. To that end, a step-by-step tutorial is presented in the Supporting Information, with notes and warnings to avoid unintentional mistakes. Regarding the levels of theory, which now can be chosen in an unlimited way, we suggest optimizing the geometries and conducting the GIAO NMR calculations at DFT levels using at least the 6-31G* and 6-31+G** basis sets (or equivalents), including the solvent effect with PCM or analogous solvation models. Making a balance between computational cost and effectiveness of results, we continue recommending the PCM/mPW1PW91/6-31+G**//B3LYP/6-31G* level for broad applications. When dealing with molecules featuring multiple intramolecular H-bonding interactions leading to unclear conformational landscapes, the parallel calculation of DP4+ using randomly generated ensembles would reinforce the confidence in the assignment.[21] If other new levels are to be used, the [μ,σ,ν] parameters should be properly estimated as discussed above and their performances should be first validated with a known set of molecules.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c00987.

Computational details, and computational data associated with this paper; DP4+ workflow, general recommendations, and instructions for using the Excel spreadsheet and tutorials (PDF)

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Notes

The authors declare no competing financial interest.

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